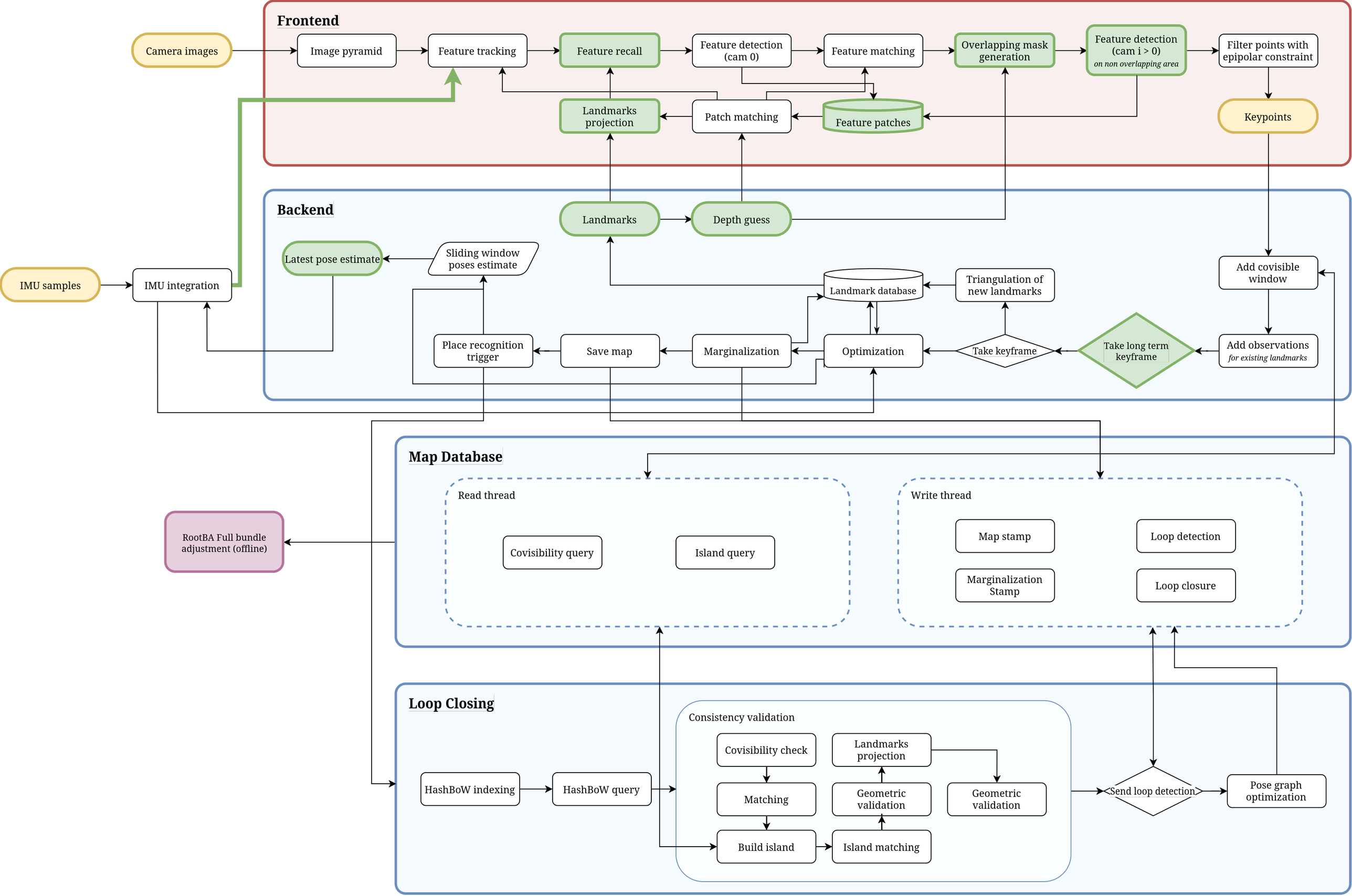

Frontend
Camera images
Image pyramid
Feature tracking
Feature recall
Feature detection (cam 0)
Feature matching
Overlapping mask generation
Feature detection (cam i > 0)
on non overlapping area
Filter points with epipolar constraint
Landmarks projection
Patch matching
Feature patches
Keypoints
Backend
Landmarks
Depth guess
Latest pose estimate
Sliding window poses estimate
IMU samples
IMU integration
Landmark database
Triangulation of new landmarks
Add covisible window
Place recognition trigger
Save map
Marginalization
Optimization
Take keyframe
Take long term keyframe
Add observations
for existing landmarks
Map Database
Read thread
Covisibility query
Island query
Write thread
Map stamp
Loop detection
Marginalization Stamp
Loop closure
RootBA Full bundle adjustment (offline)
Loop Closing
Consistency validation
HashBoW indexing
HashBoW query
Covisibility check
Matching
Build island
Landmarks projection
Geometric validation
Island matching
Geometric validation
Send loop detection
Pose graph optimization